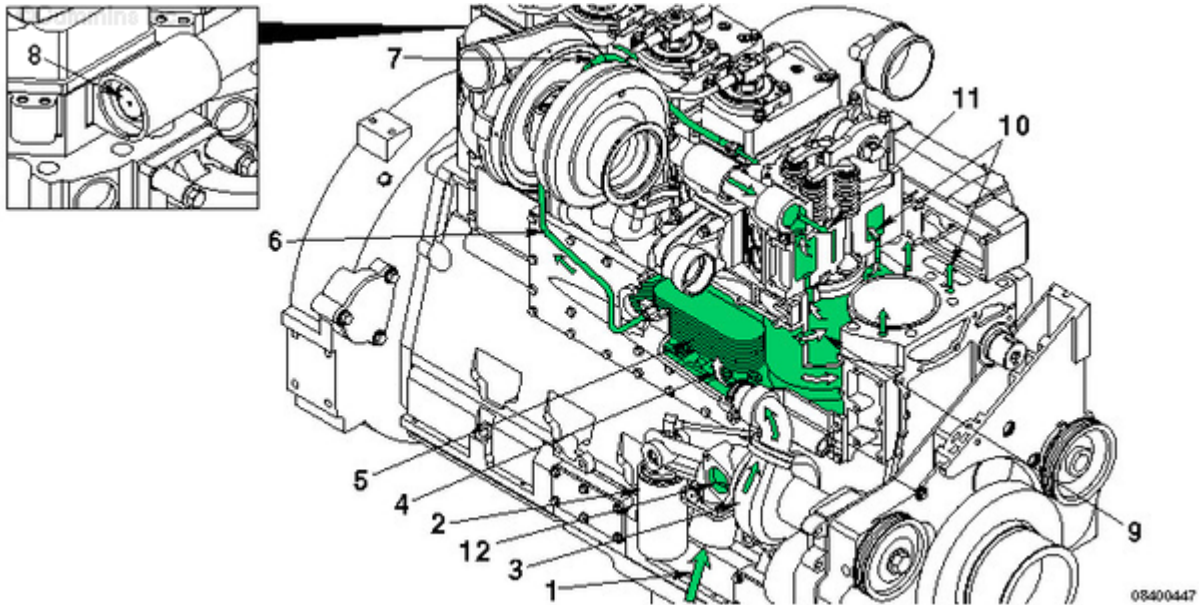


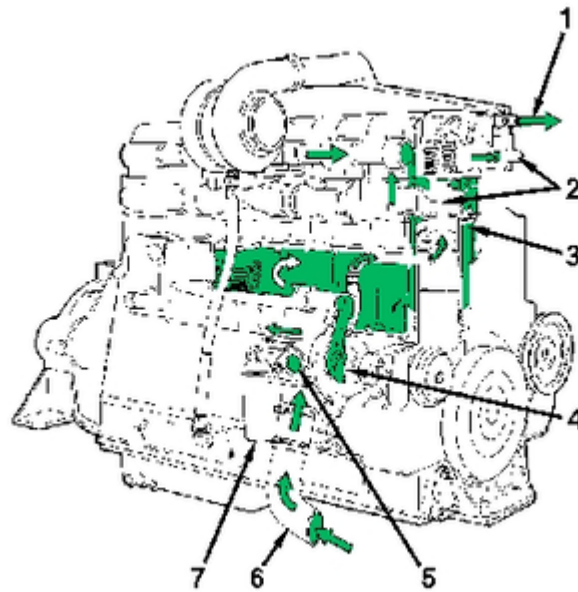
Trainings ▾

Flow Diagram



QSK19 Engine - Electronically Actuated Injectors

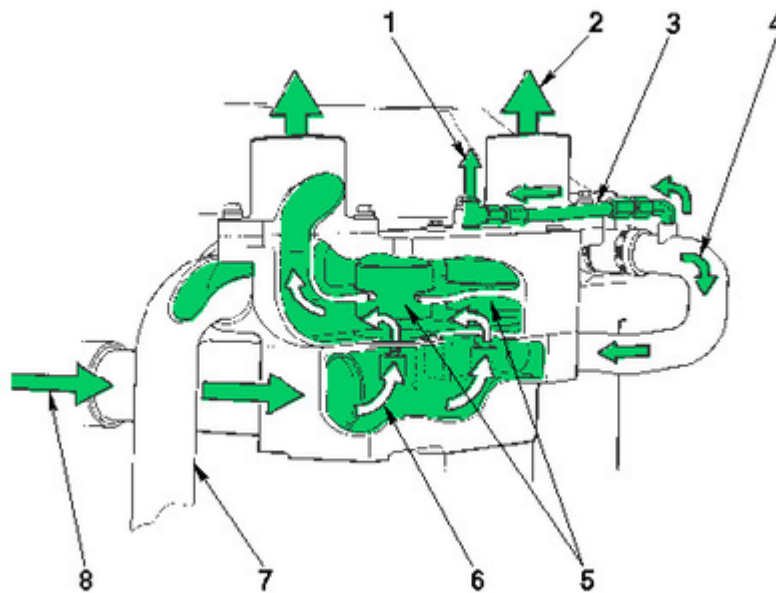
1. Coolant inlet
2. Coolant filter
3. Coolant pump
4. Coolant supply to engine block
5. Lubricating oil cooler
6. Coolant supply to turbocharger
7. Coolant return from turbocharger
8. Cabin heater port
9. Coolant flow to cylinder liners
10. Coolant flow to cylinder head
11. Coolant return to thermostat housing
12. Coolant bypass from thermostat.



09490002

QSK19 Engine With Conventional Aftercooling - Mechanically Actuated Injectors

1. Coolant out of aftercooler
2. Coolant to aftercooler
3. Cylinder liner
4. Water pump
5. Coolant bypass from thermostat
6. Water inlet
7. Water filter.

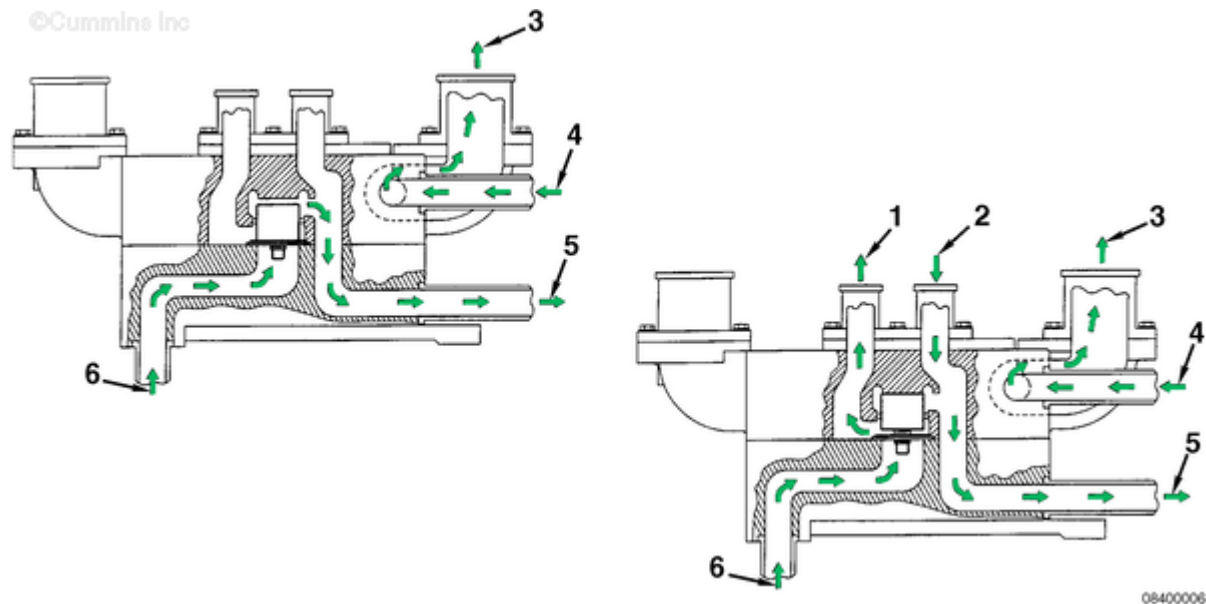


08400660

QSK Engine - Thermostat Housing

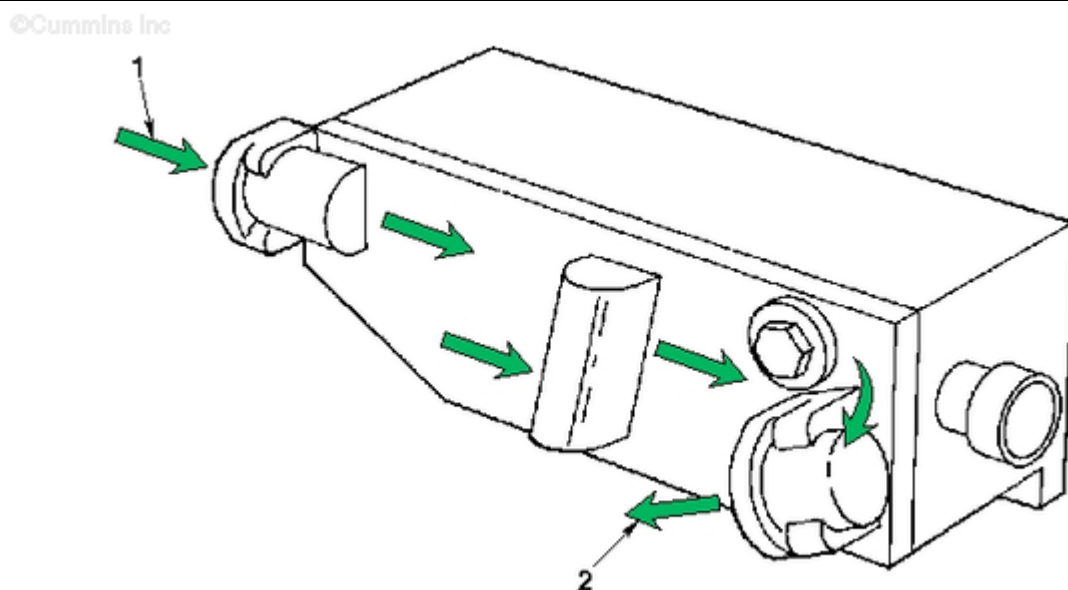
1. Vent to radiator top tank
2. Coolant to radiator
3. Vent line from aftercooler
4. Coolant out of aftercooler
5. Thermostats

6. Coolant before thermostats
7. Coolant bypass
8. Coolant manifold.



QSK19 Engines With LTA - Mechanically Actuated Injectors

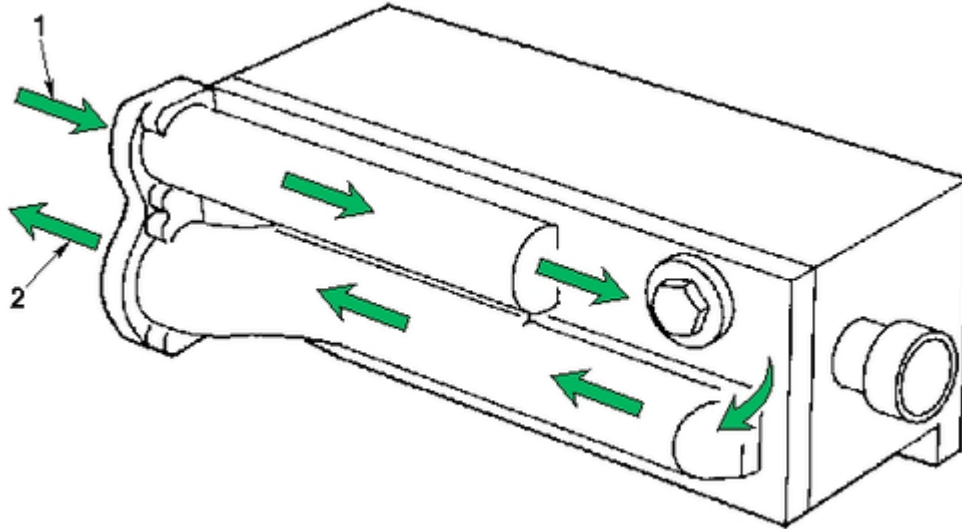
1. To aftercooler radiator
2. From aftercooler radiator
3. To radiator
4. From aftercooler
5. To aftercooler
6. From coolant pump.



QSK19 Engine - Torque Converter Cooler

1. Torque converter oil inlet (series flow)
2. Torque converter oil outlet (series flow).

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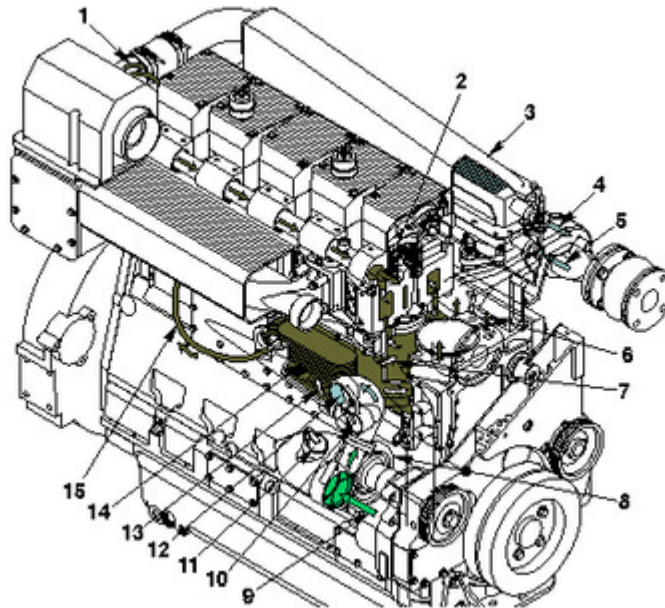


0 040 0245

QSK19 Engine - Torque Converter Cooler

1. Torque converter oil inlet (parallel flow)
2. Torque converter oil outlet (parallel flow)

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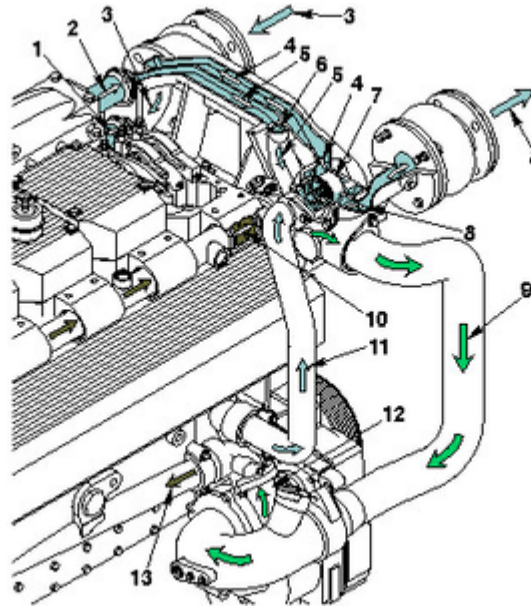
03400415

QSK19 Marine Single Loop LTA Cooling - Mechanically Actuated Injectors

1. Coolant return from turbocharger
2. Coolant return to thermostat housing
3. Aftercooler
4. Coolant return from aftercooler
5. Coolant supply to aftercooler
6. Coolant flow to cylinder head
7. Coolant flow to cylinder liners
8. Coolant pump
9. Coolant return to coolant pump

10. Emergency pump connection outlet
11. Emergency pump connection inlet
12. Coolant supply to thermostat housing
13. Coolant supply to engine block
14. Lubricating oil cooler
15. Coolant supply to turbocharger.

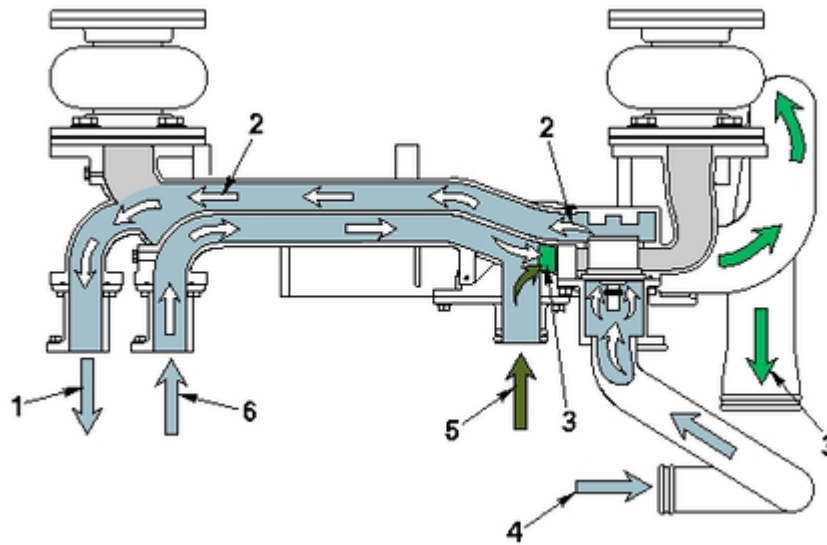
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08400414

QSK19 Marine Single Loop LTA Keel Cooling - Mechanically Actuated Injectors

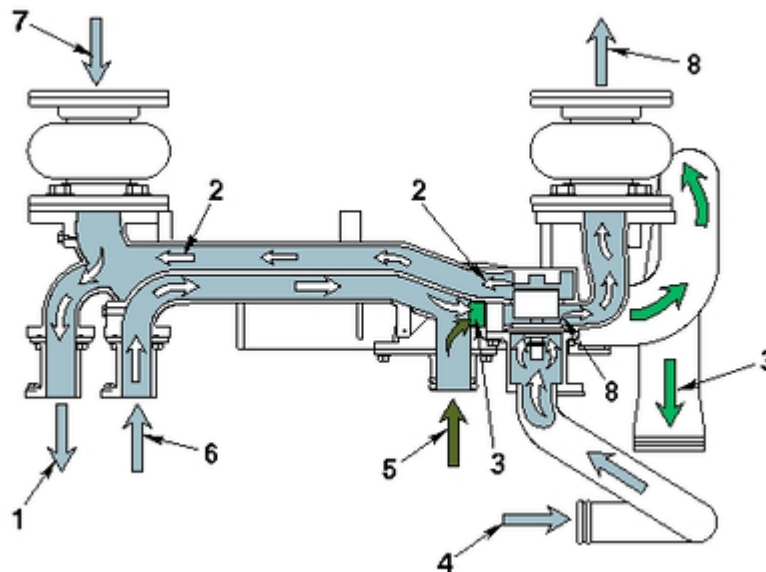
1. Aftercooler coolant supply
2. Aftercooler coolant return to thermostat housing
3. Coolant from keel cooler
4. Coolant bypass flow to aftercooler
5. Aftercooler coolant return to coolant pump
6. Coolant make-up line from expansion tank
7. Thermostat
8. Coolant to keel cooler
9. Coolant return to coolant pump
10. Coolant bypass to thermostat housing
11. Coolant supply to thermostat housing
12. Coolant pump
13. Coolant supply to engine block.



00400551

QSK19 Marine Single Loop LTA Thermostat Housing Coolant Flow, Thermostat Closed

1. Aftercooler coolant supply
2. Coolant bypass flow to aftercooler
3. Coolant return to coolant pump
4. Coolant supply from coolant pump
5. Block coolant return to thermostat housing
6. Aftercooler coolant return to thermostat housing.

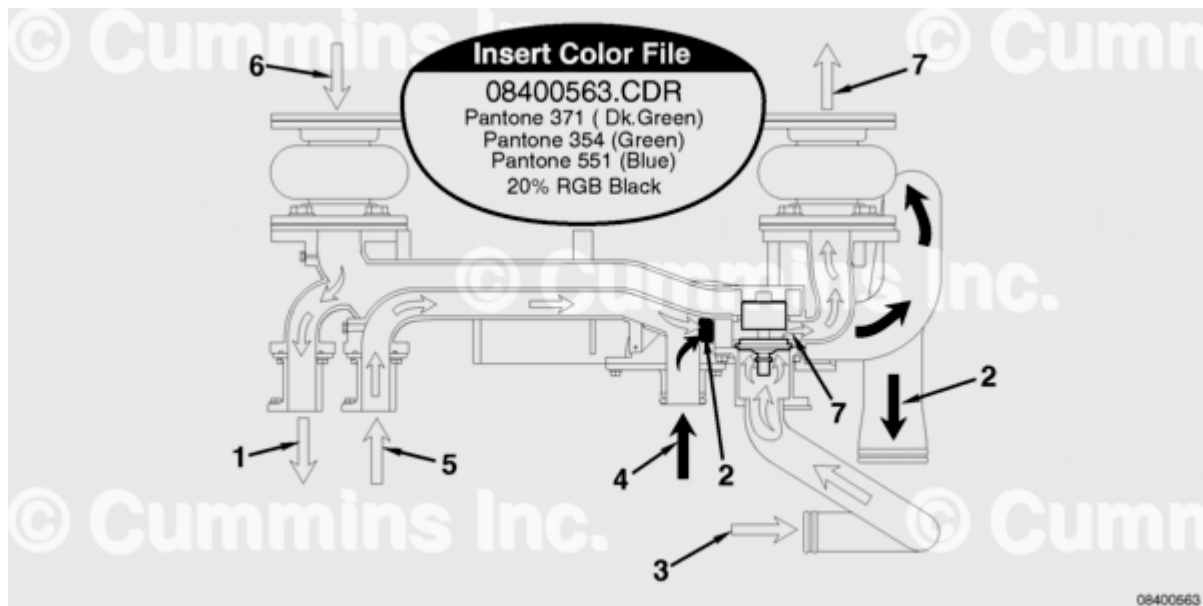


00400552

QSK19 Marine Single Loop LTA Thermostat Housing Coolant Flow, Thermostat Partially Open

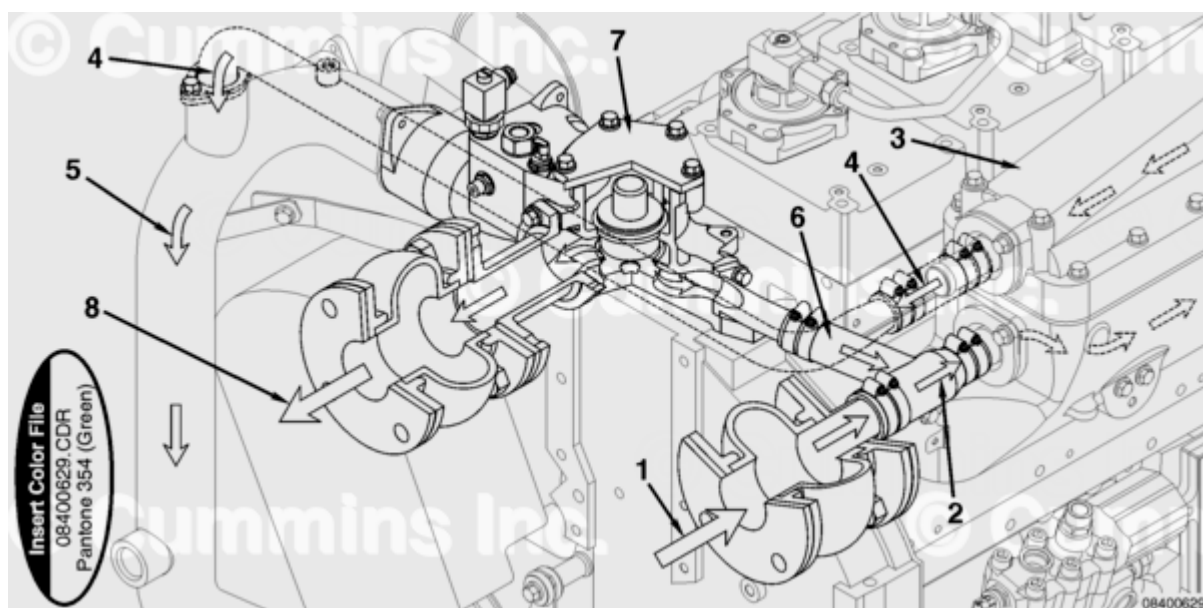
1. Aftercooler coolant supply
2. Coolant bypass flow to aftercooler
3. Coolant return to coolant pump
4. Coolant supply from coolant pump
5. Block coolant return to thermostat housing
6. Aftercooler coolant return to thermostat housing

7. Coolant from keel cooler
8. Coolant to keel cooler.



QSK19 Marine Single Loop LTA Thermostat Housing Coolant Flow, Thermostat Open

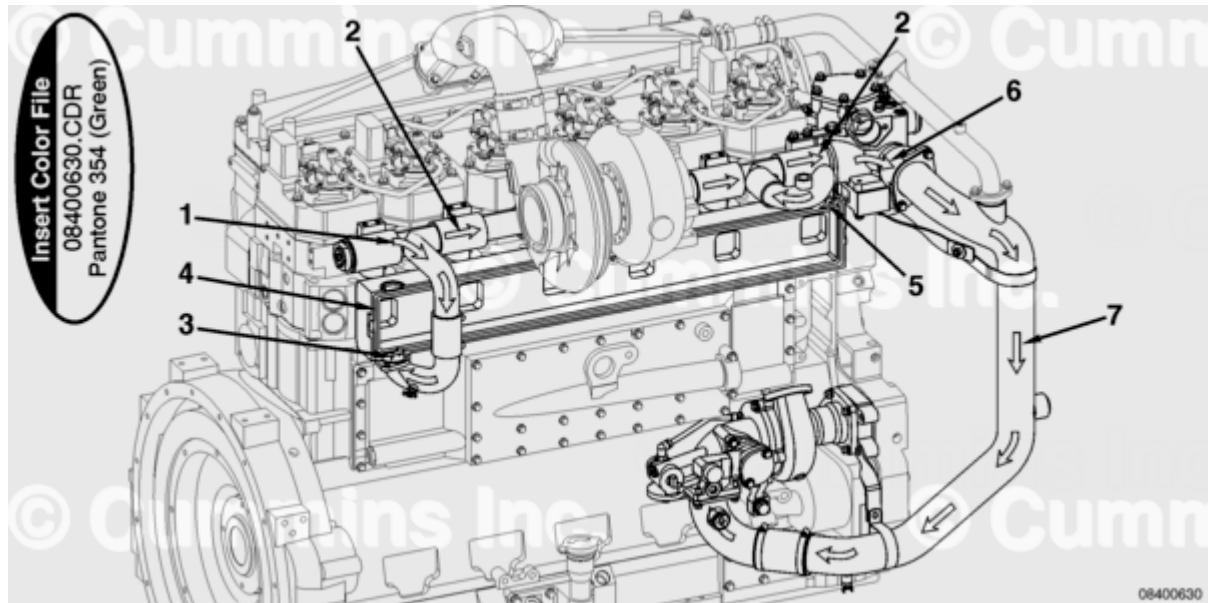
1. Aftercooler coolant supply
2. Coolant return to coolant pump
3. Coolant supply from coolant pump
4. Block coolant return to thermostat housing
5. Aftercooler coolant return to thermostat housing
6. Coolant from keel cooler
7. Coolant to keel cooler.



QSK19 Engine, Keel Cooled - Electronically Actuated Injectors

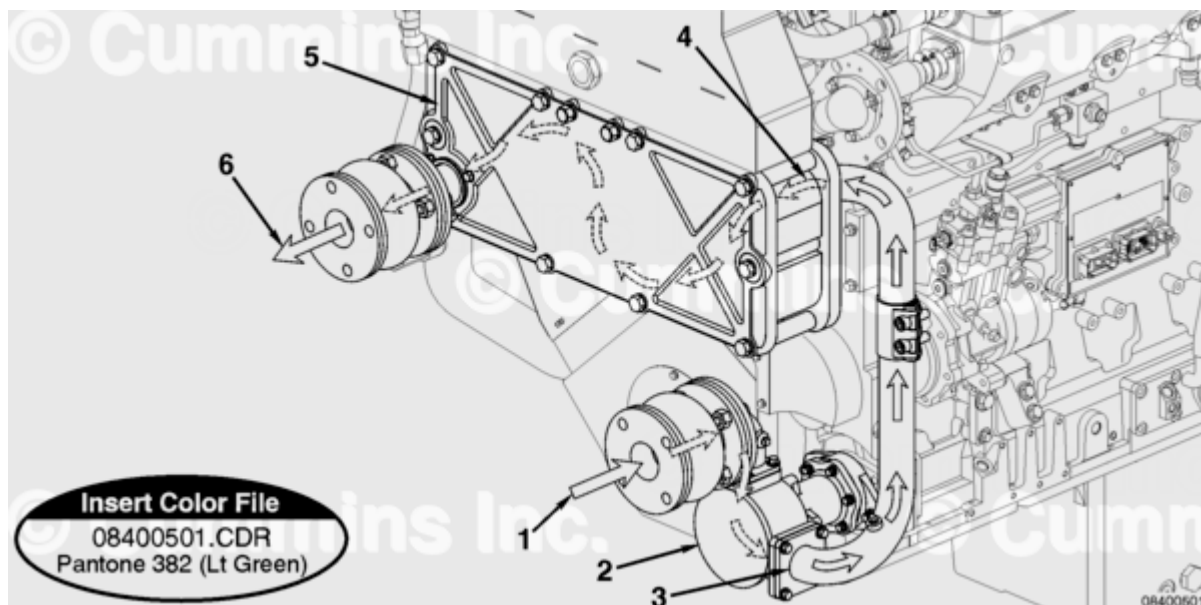
1. Coolant flow from keel cooler
2. Coolant flow to aftercooler

3. Aftercooler
4. Coolant flow from aftercooler to water pump return
5. Engine water pump return
6. Coolant flow from the thermostat (partially open/closed)
7. Thermostat
8. Coolant return to keel cooler.



QSK19 Marine Exhaust Coolant Flow - Electronically Actuated Injectors

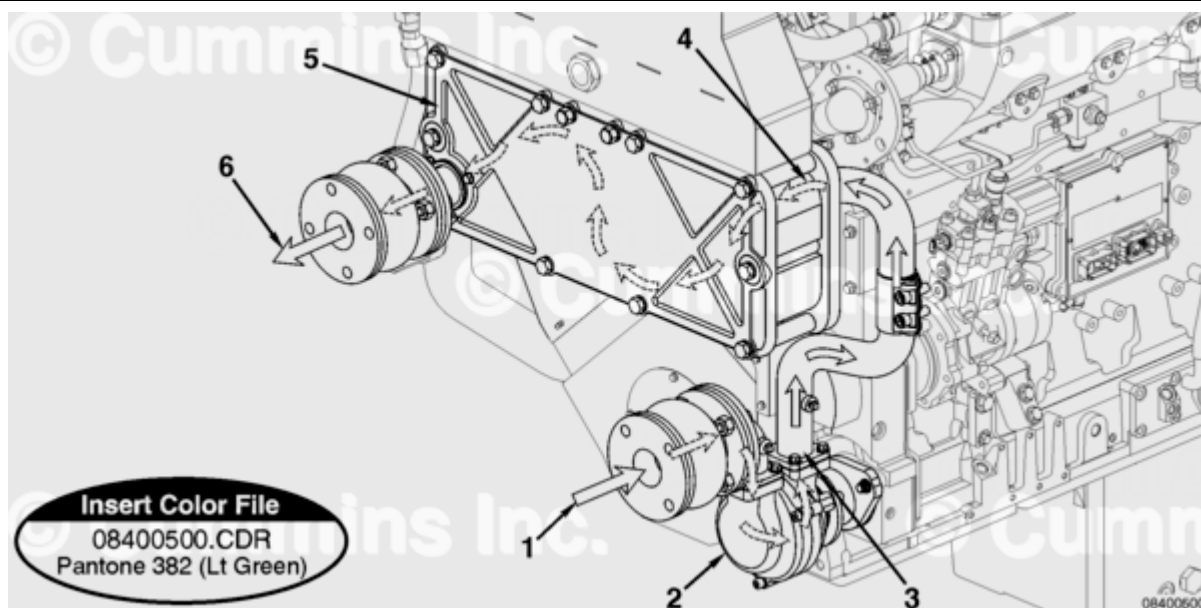
1. Coolant flow to exhaust manifold
2. Coolant flow to thermostat housing
3. Coolant flow into exhaust manifold
4. Coolant flow through exhaust manifold
5. Coolant flow out of exhaust manifold
6. Coolant flow from thermostat housing to water pump return
7. Coolant return to water pump inlet.



QSK19 Engine, Heat Exchanger - Electronically Actuated Injector

1. Sea water inlet
2. Sea water pump
3. Sea water pump outlet to heat exchanger
4. Sea water inlet to heat exchanger
5. Heat exchanger
6. Sea water outlet.

Note : This application of the heat exchanger uses a Jabsco™ water pump.



QSK19 Engine, Heat Exchanger - Electronically Actuated Injector

1. Sea water inlet
2. Sea water pump
3. Sea water pump outlet to heat exchanger
4. Sea water inlet to heat exchanger
5. Heat exchanger

6. Sea water outlet.

Note : This application of the Marine heat exchanger uses a Gilkes™ water pump.

Last Modified: 10-Dec-2013
